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Roll No. 11

T.Y A(AIML)

✓ Mini Assignment — Full EDA on the Penguins Dataset

Perform a complete EDA on the `penguins` dataset (already loaded) and answer the following in new cells below. This is meant to be done **independently** — use everything you've learned in this notebook!

1. Number of rows and columns
2. Number of missing values (per column)
3. Number of duplicate rows
4. A correlation matrix / heatmap of the numeric columns
5. A boxplot showing `bill_length_mm` grouped by `species`
6. A pairplot colored by `species`
7. A fully cleaned version of the dataset (missing values handled, duplicates removed)

Use the empty code cell below to get started:

```
import seaborn as sns
penguins = sns.load_dataset('penguins')

# Your Mini Assignment code goes here
penguins.head()
```

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_
0	Adelie	Torgersen	39.1	18.7	181.0	
1	Adelie	Torgersen	39.5	17.4	186.0	
2	Adelie	Torgersen	40.3	18.0	195.0	
3	Adelie	Torgersen	NaN	NaN	NaN	
4	Adelie	Torgersen	36.7	19.3	193.0	

1. Number of rows and columns

```
# Number of rows and columns
print("Shape of Dataset:", penguins.shape)
print("Rows:", penguins.shape[0])
print("Columns:", penguins.shape[1])
```

```
Shape of Dataset: (344, 7)
Rows: 344
Columns: 7
```

2. Number of missing values (per column)

```
# Missing values per column
print("Missing Values:")
print(penguins.isnull().sum())
```

```
Missing Values:
species          0
island           0
bill_length_mm   2
bill_depth_mm    2
flipper_length_mm 2
body_mass_g      2
sex             11
dtype: int64
```

3. Number of duplicate rows

```
# Number of duplicate rows
duplicates = penguins.duplicated().sum()
print("Duplicate Rows:", duplicates)
```

```
Duplicate Rows: 0
```

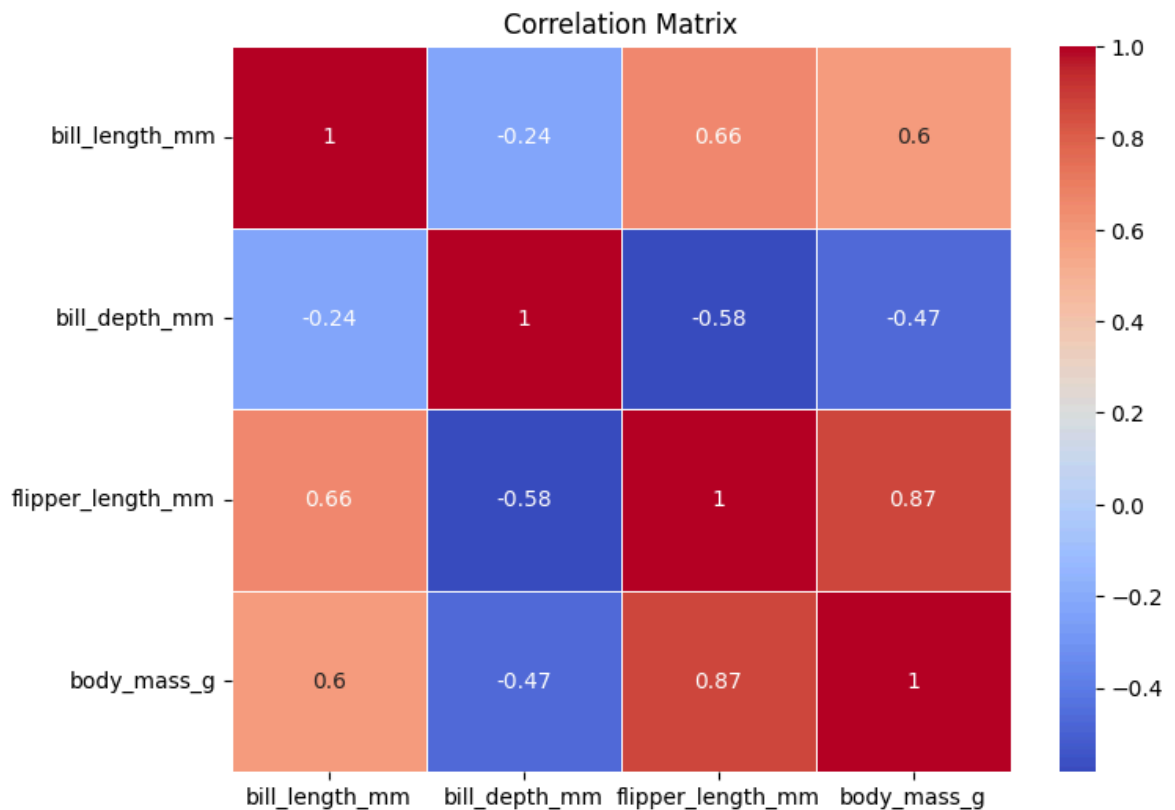
4. Correlation matrix and heatmap

```
import matplotlib.pyplot as plt
import seaborn as sns

# Select only numeric columns
numeric_data = penguins.select_dtypes(include='number')

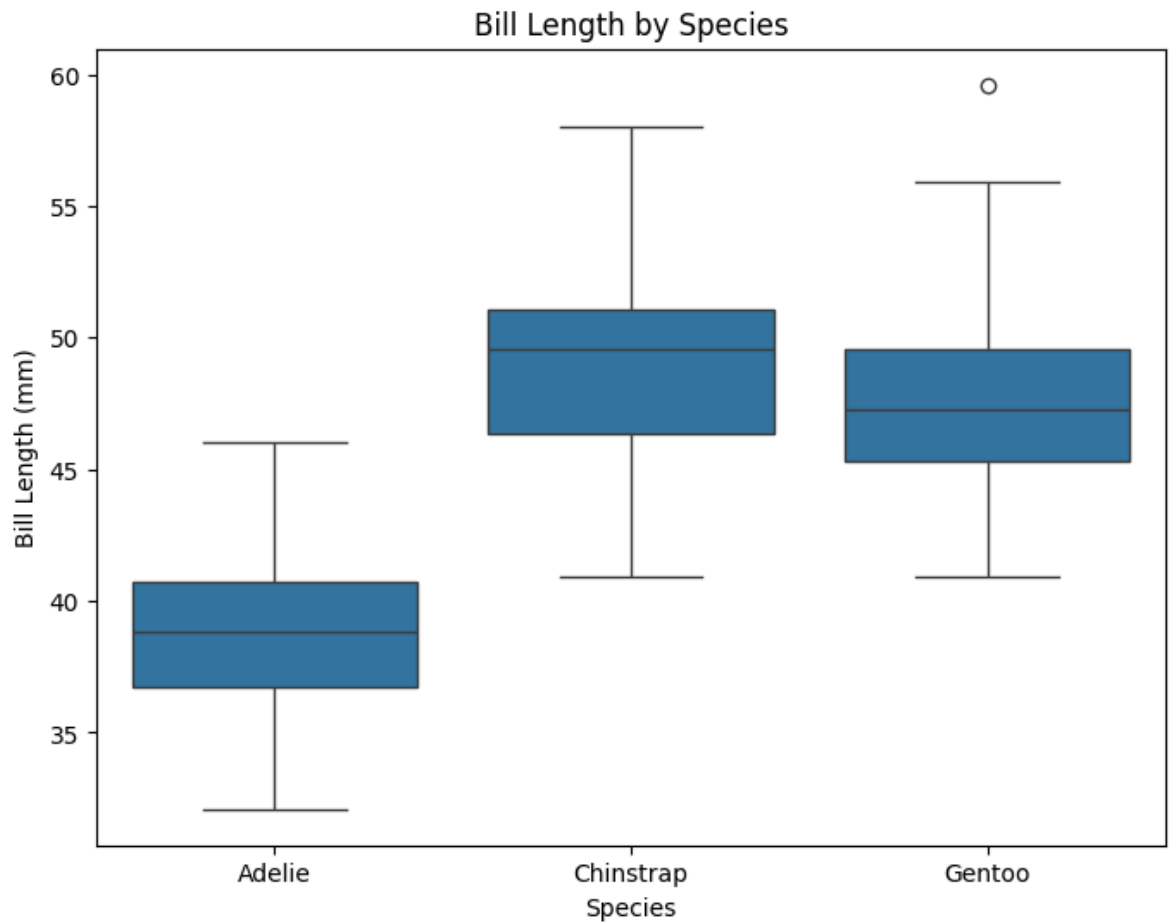
# Correlation matrix
corr_matrix = numeric_data.corr()
```

```
# Heatmap
plt.figure(figsize=(8,6))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', linewidths=0.5)
plt.title("Correlation Matrix")
plt.show()
```



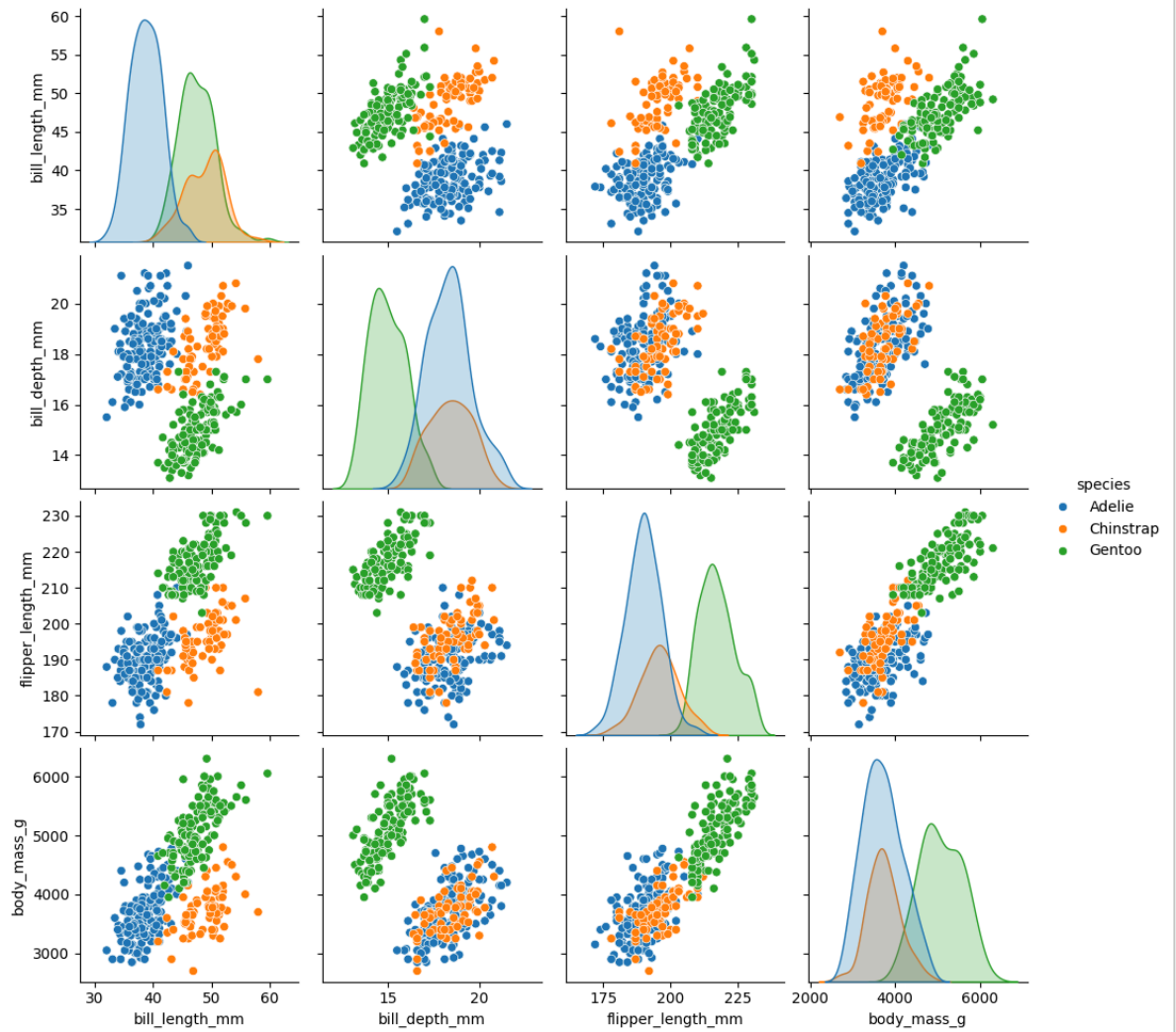
✓ 5. Boxplot of bill_length_mm grouped by species

```
plt.figure(figsize=(8,6))
sns.boxplot(data=penguins, x="species", y="bill_length_mm")
plt.title("Bill Length by Species")
plt.xlabel("Species")
plt.ylabel("Bill Length (mm)")
plt.show()
```



✓ 6. Pairplot colored by species

```
sns.pairplot(penguins, hue="species")  
plt.show()
```



```
# Remove duplicate rows
penguins_clean = penguins.drop_duplicates()

# Handle missing values by removing rows with missing data
penguins_clean = penguins_clean.dropna()

print("Original Shape:", penguins.shape)
print("Cleaned Shape:", penguins_clean.shape)

# Display first few rows
penguins_clean.head()
```

Original Shape: (344, 7)
Cleaned Shape: (333, 7)

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
0	Adelie	Torgersen	39.1	18.7	181.0	3750
1	Adelie	Torgersen	39.5	17.4	186.0	3800
2	Adelie	Torgersen	40.3	18.0	195.0	3950
4	Adelie	Torgersen	36.7	19.3	193.0	3650
5	Adelie	Torgersen	39.3	20.6	190.0	3800

Next steps: [Generate code with penguins_clean](#)

[New interactive sheet](#)

✓ Bonus: Verify the cleaned dataset

```
print("Missing Values After Cleaning:")
print(penguins_clean.isnull().sum())

print("\nDuplicate Rows After Cleaning:")
print(penguins_clean.duplicated().sum())
```

Missing Values After Cleaning:

species	0
island	0
bill_length_mm	0
bill_depth_mm	0
flipper_length_mm	0